Lecture 21

Smooth Vectors, K-Types and $(\mathfrak{g}, K)$-Modules

Objectives. *A representation of a noncompact group usually acts on an infinite-dimensional Hilbert space. We use its derivatives on smooth vectors. These operators are in general unbounded, and the space of smooth vectors has no convenient basis. Let K be a compact subgroup. If $\Pi|_K$ is unitary, then Lecture 10 splits the Hilbert space into a Hilbert sum of blocks, one for each K-type. So we pass to the smooth vectors that are finite under K. On these vectors $\mathfrak{g}$ and K act together, subject to compatibility conditions. For $\mathrm{SL}(2,\mathbb{R})$ with $K = \mathrm{SO}(2)$, the study of a representation becomes a raising and lowering calculation.*

By the end of this lecture you will be able to:

- Define smooth vectors, K-finite vectors, K-types and admissibility.
- Prove that the K-finite smooth vectors of a strongly continuous representation on a Hilbert space form a $(\mathfrak{g}, K)$-module.
- Use the compatibility conditions to show that E and F shift the circle K-types of $\mathrm{SL}(2,\mathbb{R})$ by ± 2.
- State which global questions the $(\mathfrak{g}, K)$-module leaves open, and which cited theorems address them.

21.1 Smooth vectors and K-finite vectors

Throughout, G is a Lie group with Lie algebra $\mathfrak{g}$. The subgroup $K \subseteq G$ is compact, with Lie algebra $\mathfrak{k} \subseteq \mathfrak{g}$. The representation Π of G on a Hilbert space $\mathcal{H}$ is strongly continuous. Here each operator $\Pi(g)$ is bounded, linear and invertible. The map $g \mapsto \Pi(g)v$ from G to $\mathcal{H}$ is continuous for every $v \in \mathcal{H}$. Unitarity is assumed only where stated.

Definition 21.1. A vector $v \in \mathcal{H}$ is *smooth* if the orbit map $G \to \mathcal{H}$, $g \mapsto \Pi(g)v$, is smooth. This means that in every chart all iterated partial derivatives exist as norm limits and are continuous. We write $\mathcal{H}^\infty$ for the set of smooth vectors. For $v \in \mathcal{H}^\infty$ and $X \in \mathfrak{g}$ we set

$$d\Pi(X)v = \left.\frac{d}{dt}\right|_{t=0} \Pi(\exp tX)v.$$

The set $\mathcal{H}^\infty$ is a subspace of $\mathcal{H}$, because a linear combination of smooth orbit maps is smooth.

For unitary representations, the following facts were imported in Lecture 15. We now also import the density and invariance facts (F1)–(F2) for strongly continuous representations by bounded operators. A source for this more general smooth-vector theorem is Knapp RT, Sections III.3–4. Here Knapp RT is A. W. Knapp, *Representation Theory of Semisimple Groups: An Overview Based on Examples* (2001).

(F1) *Density* (Gårding). The space $\mathcal{H}^\infty$ is dense in $\mathcal{H}$.

(F2) *Invariance.* Each operator $d\Pi(X)$ maps $\mathcal{H}^\infty$ into itself, and $X \mapsto d\Pi(X)$ is a Lie algebra representation of $\mathfrak{g}$ on $\mathcal{H}^\infty$.

(F3) *Skew-symmetry.* If Π is unitary, then $\langle d\Pi(X)u, w\rangle = -\langle u, d\Pi(X)w\rangle$ for all $X \in \mathfrak{g}$ and $u, w \in \mathcal{H}^\infty$.

We extend $d\Pi$ complex-linearly to $\mathfrak{g}_{\mathbb{C}}$ by $d\Pi(X + iY) = d\Pi(X) + i\, d\Pi(Y)$ for $X, Y \in \mathfrak{g}$. By the invariance property (F2), this is a representation of $\mathfrak{g}_{\mathbb{C}}$ on $\mathcal{H}^\infty$.

One elementary fact is proved here. If $v \in \mathcal{H}^\infty$ and $h \in G$, then $\Pi(h)v \in \mathcal{H}^\infty$. Indeed, the orbit map of $\Pi(h)v$ is $g \mapsto \Pi(gh)v$. This is the orbit map of v composed with the smooth right translation $g \mapsto gh$.

Definition 21.2. A vector $v \in \mathcal{H}$ is *K-finite* if $\operatorname{span}\{\Pi(k)v : k \in K\}$ is finite-dimensional. We write $\mathcal{H}_K$ for the set of K-finite vectors.

The set $\mathcal{H}_K$ is a subspace, because for $a, b \in \mathbb{C}$ the span of the vectors $\Pi(k)(av + bw)$, $k \in K$, lies in $\operatorname{span}\Pi(K)v + \operatorname{span}\Pi(K)w$.

Let $\widehat{K}$ be the set of equivalence classes of irreducible continuous representations of K. Each class δ has a finite-dimensional representative space V_δ (Lecture 10). Let $\operatorname{Hom}_K(V_\delta, \mathcal{H})$ be the space of linear maps $V_\delta \to \mathcal{H}$ that commute with K.

Definition 21.3. The *multiplicity* of $\delta \in \widehat{K}$ in Π is $m_\delta(\Pi) = \dim \operatorname{Hom}_K(V_\delta, \mathcal{H})$. The class δ is a *K-type* of Π if $m_\delta(\Pi) \geq 1$. The representation Π is *admissible* if $m_\delta(\Pi) < \infty$ for every $\delta \in \widehat{K}$.

Let $v \in \mathcal{H}_K$ and $W_v = \operatorname{span}\Pi(K)v$. Then W_v is finite-dimensional and K-invariant. All norms on W_v are equivalent, so strong continuity makes K act continuously on W_v. By complete reducibility (Lecture 5), W_v is a direct sum of irreducible subspaces. Hence every K-finite vector is a finite sum of vectors from K-types.

For the circle we write k_θ, $\theta \in \mathbb{R}/2\pi\mathbb{Z}$, with $k_\theta k_\phi = k_{\theta+\phi}$. Its irreducible continuous representations are the characters $\chi_n(k_\theta) = e^{in\theta}$, $n \in \mathbb{Z}$ (Lecture 7). We label the K-type χ_n by n.

Example 21.4. Let $G = K = \mathrm{SO}(2)$ and $\mathcal{H} = L^2(\mathbb{R}/2\pi\mathbb{Z})$, with

$$\langle f, h\rangle = \frac{1}{2\pi}\int_0^{2\pi} \overline{f(x)}\, h(x)\, dx, \qquad (\Pi(k_\theta)f)(x) = f(x + \theta).$$

This representation is unitary and strongly continuous. The functions $e_n(x) = e^{inx}$ form an orthonormal basis (Lecture 10). Put $\mathcal{H}(n) = \{v : \Pi(k_\theta)v = e^{in\theta}v \text{ for all } \theta\}$. Then $e_n \in \mathcal{H}(n)$.

We show that $\mathcal{H}(n) = \mathbb{C}e_n$. Let $a_m(v) = \langle e_m, v\rangle$. For continuous v, the substitution $y = x + \theta$ and periodicity give

$$a_m(\Pi(k_\theta)v) = \frac{1}{2\pi}\int_0^{2\pi} e^{-im(y-\theta)}v(y)\,dy = e^{im\theta}a_m(v).$$

Both sides are continuous linear functionals of v, so density extends the identity to all L^2 vectors. If $v \in \mathcal{H}(n)$, the left side is also $e^{in\theta}a_m(v)$. For $m \neq n$ take $\theta = \pi/(m-n)$. Then $e^{im\theta} = -e^{in\theta}$, so $a_m(v) = 0$. Hence $v = a_n(v)e_n$.

So $\mathcal{H}_K$ is the space of trigonometric polynomials $f = \sum_{|n|\le N} c_n e_n$. Each such f is smooth, because its orbit map

$$\theta \mapsto \Pi(k_\theta)f = \sum_{|n|\le N} c_n\, e^{in\theta}\, e_n$$

is a finite sum of smooth scalar functions times fixed vectors. Thus every K-finite vector is smooth. Every multiplicity is 1, so Π is admissible. The space $\mathcal{H}_K$ is dense but proper. For example, $\sum_n 2^{-|n|}e_n$ is not K-finite.

A K-finite vector need not be smooth. Let $G = \mathbb{R}$ act on $L^2(\mathbb{R})$ by translation, and let $K = \{0\}$. Every vector is K-finite. The indicator function of $[0,1]$ is not smooth (Exercise 21.5). The domain example of Lecture 15 computes the same difference quotients. So we impose both conditions. We write $\mathcal{H}_K^\infty$ for the space of smooth K-finite vectors:

$$\mathcal{H}_K^\infty = \mathcal{H}^\infty \cap \mathcal{H}_K.$$

21.2 The compatibility conditions

Definition 21.5. A $(\mathfrak{g}, K)$-*module* is a complex vector space V with a Lie algebra representation of $\mathfrak{g}$ on V, written $v \mapsto X \cdot v$, and a group homomorphism $K \to \mathrm{GL}(V)$, written $v \mapsto k \cdot v$, such that:

(a) (*finiteness condition*) every $v \in V$ lies in a finite-dimensional K-invariant subspace W on which K acts continuously;

(b) (*derivative condition*) for $X \in \mathfrak{k}$ and $v \in V$, with W as in the finiteness condition (a),

$$X \cdot v = \left.\frac{d}{dt}\right|_{t=0} \exp(tX)\cdot v \qquad \text{(derivative computed in } W\text{)};$$

(c) (*covariance condition*) for $k \in K$, $X \in \mathfrak{g}$ and $v \in V$,

$$k \cdot (X \cdot v) = (\mathrm{Ad}(k)X) \cdot (k \cdot v).$$

A *morphism* of $(\mathfrak{g}, K)$-modules is a linear map that commutes with both actions. A $(\mathfrak{g}, K)$-module V is *admissible* if $\dim \mathrm{Hom}_K(V_\delta, V) < \infty$ for every $\delta \in \widehat{K}$.

The derivative in the derivative condition (b) does not depend on the choice of W. Indeed, two choices W and W' both lie in the finite-dimensional space $W + W'$, and all norms on $W + W'$ are equivalent.

The $\mathfrak{g}$-action extends complex-linearly to $\mathfrak{g}_\mathbb{C}$. Both sides of the covariance condition (c) are complex-linear in X, so the covariance condition also holds for $X \in \mathfrak{g}_\mathbb{C}$.

The derivative condition (b) involves only elements $\exp(tX)$, which lie in the identity component of K. Hence, if K is disconnected, the K-action carries information that $\mathfrak{k}$ cannot detect (Exercise 21.7).

The covariance condition (c) describes how the K-action transforms the $\mathfrak{g}$-action. In § 21.3 it gives the shifts of the circle K-types for $\mathrm{SL}(2,\mathbb{R})$.

Proposition 21.6. *Let Π be a strongly continuous representation of G on $\mathcal{H}$, and let $K \subseteq G$ be compact. Then $\mathcal{H}_K^\infty$ is invariant under $\Pi(k)$ for $k \in K$ and under $d\Pi(X)$ for $X \in \mathfrak{g}_{\mathbb{C}}$. With these actions $\mathcal{H}_K^\infty$ is a $(\mathfrak{g}, K)$-module. Let Π' be a strongly continuous representation of G on a Hilbert space $\mathcal{H}'$. Let $T : \mathcal{H} \to \mathcal{H}'$ be bounded and linear, with $T\Pi(g) = \Pi'(g)T$ for all $g \in G$. Then T maps $\mathcal{H}_K^\infty$ into $\mathcal{H}_K'^\infty$, and its restriction is a morphism.*

Proof. Step 1: the K-action. Let $v \in \mathcal{H}_K^\infty$ and $k \in K$. Then $\Pi(k)v$ is smooth by the right-translation argument in § 21.1. Since $Kk = K$,

$$\operatorname{span} \Pi(K)\Pi(k)v = \operatorname{span} \Pi(Kk)v = \operatorname{span} \Pi(K)v,$$

which is finite-dimensional. So $\Pi(k)v \in \mathcal{H}_K^\infty$. In particular, $W_v = \operatorname{span} \Pi(K)v$ is contained in $\mathcal{H}_K^\infty$.

Step 2: the covariance identity. Let $k \in K$, $X \in \mathfrak{g}$ and $w \in \mathcal{H}^\infty$. We claim

$$\Pi(k)\, d\Pi(X)w = d\Pi(\mathrm{Ad}(k)X)\, \Pi(k)w. \tag{21.1}$$

By the conjugation formula for exponentials (Lecture 2), $k \exp(tX)k^{-1} = \exp(t\,\mathrm{Ad}(k)X)$. Hence, for $t \neq 0$,

$$\Pi(k)\Big[\frac{\Pi(\exp tX)w - w}{t}\Big] = \frac{\Pi(\exp(t\,\mathrm{Ad}(k)X))\,\Pi(k)w - \Pi(k)w}{t}.$$

Let $t \to 0$. On the left, the bracket tends to $d\Pi(X)w$, and $\Pi(k)$ is bounded. So the left side tends to $\Pi(k)\, d\Pi(X)w$. On the right, $\Pi(k)w$ is smooth, so the quotient tends to $d\Pi(\mathrm{Ad}(k)X)\Pi(k)w$.

Step 3: $\mathfrak{g}$ preserves K-finiteness. This is the main step. The difficulty is that $d\Pi(X)$ does not commute with $\Pi(k)$. So the span of $\Pi(K)\, d\Pi(X)v$ is not obviously finite-dimensional. The covariance identity (21.1) replaces commutation. It moves X to $\mathrm{Ad}(k)X$, which still lies in the finite-dimensional space $\mathfrak{g}$.

Let $v \in \mathcal{H}_K^\infty$ and $W = W_v$. By Step 1, $W \subseteq \mathcal{H}^\infty$, so $d\Pi(X)w$ is defined for $w \in W$. Choose bases $X_1, \dots, X_d$ of $\mathfrak{g}$ and $w_1, \dots, w_r$ of W. Put

$$U = \operatorname{span}\{d\Pi(X_i)w_j : 1 \le i \le d,\ 1 \le j \le r\}.$$

Then $\dim U \le dr$. Let $X \in \mathfrak{g}$, $w \in W$ and $k \in K$. Expand

$$\mathrm{Ad}(k)X = \sum_{i=1}^{d} a_i(k)X_i, \qquad \Pi(k)w = \sum_{j=1}^{r} b_j(k)w_j,$$

with scalars $a_i(k)$ and $b_j(k)$. The second expansion uses $\Pi(k)w \in W$. By the covariance identity (21.1) and linearity of $d\Pi$ in both arguments,

$$\Pi(k)\, d\Pi(X)w = d\Pi(\mathrm{Ad}(k)X)\, \Pi(k)w = \sum_{i=1}^{d}\sum_{j=1}^{r} a_i(k)\, b_j(k)\, d\Pi(X_i)w_j \in U.$$

In particular $\Pi(K)\, d\Pi(X)v \subseteq U$. So $d\Pi(X)v$ is K-finite. It is smooth by the invariance property (F2). Complex combinations stay in the subspace $\mathcal{H}_K^\infty$, so the same holds for $X \in \mathfrak{g}_{\mathbb{C}}$.

Step 4: the axioms. By the invariance property (F2) and Step 3, $d\Pi$ is a representation of $\mathfrak{g}$ on $\mathcal{H}_K^\infty$. The finiteness condition (a) holds with $W = W_v$, by Step 1 and the continuity of the K-action on W_v shown in § 21.1. For the derivative condition (b), let $X \in \mathfrak{k}$. The curve $t \mapsto \Pi(\exp tX)v$ lies in W_v. Its derivative in $\mathcal{H}$ is $d\Pi(X)v$. A finite-dimensional subspace is closed, and W_v carries the restricted norm. So the difference quotients converge in W_v to $d\Pi(X)v$. The covariance condition (c) is the covariance identity (21.1).

Step 5: intertwiners. Let $v \in \mathcal{H}_K^\infty$. The orbit map $g \mapsto \Pi'(g)Tv = T\Pi(g)v$ is smooth, because T is bounded and linear. Also $\operatorname{span} \Pi'(K)Tv = T(\operatorname{span} \Pi(K)v)$ is finite-dimensional. Differentiating $T\Pi(\exp tX)v = \Pi'(\exp tX)Tv$ at $t = 0$ gives $T\, d\Pi(X)v = d\Pi'(X)Tv$. Commutation with K is the hypothesis. ∎

Definition 21.7. The $(\mathfrak{g}, K)$-*module of* Π is $\mathcal{H}_K^\infty$ with the actions given in the $(\mathfrak{g}, K)$-module proposition, Proposition 21.6. Two representations are *infinitesimally equivalent* if their $(\mathfrak{g}, K)$-modules are isomorphic.

21.3 Circle K-types for $\mathrm{SL}(2, \mathbb{R})$

Now $G = \mathrm{SL}(2, \mathbb{R})$ and $K = \mathrm{SO}(2)$. Put

$$W = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad k_\theta = \exp(\theta W) = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}.$$

The matrix k_θ here is $k_{-\theta}$ in the notation of Lecture 20. Then $\mathfrak{k} = \mathbb{R}W$, and $\theta \mapsto k_\theta$ identifies $\mathbb{R}/2\pi\mathbb{Z}$ with K. So K is connected.

Let V be a $(\mathfrak{g}, K)$-module. For $n \in \mathbb{Z}$ put

$$V(n) = \{v \in V : k_\theta \cdot v = e^{in\theta} v \text{ for all } \theta\}.$$

By the finiteness condition (a) and complete reducibility (Lecture 5), every $v \in V$ is a finite sum of vectors from the spaces $V(n)$. The sum is direct, because the functions $e^{in\theta}$ are linearly independent. So $V = \bigoplus_{n\in\mathbb{Z}} V(n)$, and $\dim V(n)$ is the multiplicity of the K-type n.

Let $H = \operatorname{diag}(1, -1)$ and $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then H, S, W is a basis of $\mathfrak{g} = \mathfrak{sl}(2, \mathbb{R})$. Multiplying matrices gives

$$[H, S] = 2W, \qquad [W, H] = -2S, \qquad [W, S] = 2H.$$

In $\mathfrak{g}_{\mathbb{C}} = \mathfrak{sl}(2, \mathbb{C})$ we set

$$H' = -iW = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad E = \frac{1}{2}(H + iS) = \frac{1}{2}\begin{pmatrix} 1 & i \\ i & -1 \end{pmatrix}, \quad F = \frac{1}{2}(H - iS) = \frac{1}{2}\begin{pmatrix} 1 & -i \\ -i & -1 \end{pmatrix}.$$

The real brackets give

$$\begin{aligned} [H', E] &= -\tfrac{i}{2}\big([W, H] + i[W, S]\big) = -\tfrac{i}{2}(-2S + 2iH) = H + iS = 2E, \\ [H', F] &= -\tfrac{i}{2}\big([W, H] - i[W, S]\big) = -\tfrac{i}{2}(-2S - 2iH) = -(H - iS) = -2F, \\ [E, F] &= \tfrac{1}{4}\big(-i[H, S] + i[S, H]\big) = -\tfrac{i}{2}[H, S] = -iW = H'. \end{aligned}$$

Thus (H', E, F) is an $\mathfrak{sl}_2$-triple. The $\mathfrak{sl}_2$ theory of the prerequisite Lie Algebras course applies to it. The element H' spans $\mathfrak{k}_{\mathbb{C}}$, whereas $H \notin \mathfrak{k}$. Also $F = \overline{E}$, where $\overline{X + iY} = X - iY$ for $X, Y \in \mathfrak{g}$.

These E and F are not the real matrices of Lecture 20. Let $u_{\pm} = (1, \pm i)^T$, and let Q be the matrix with columns u_+ and u_-. Then

$$H' = Q \operatorname{diag}(1, -1) Q^{-1}, \qquad E = Q \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} Q^{-1}, \qquad F = Q \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} Q^{-1}.$$

The vectors $u_{\pm}$ are eigenvectors of K, as Example 21.10 shows. In the K-eigenbasis (u_+, u_-), the matrices of (H', E, F) are the standard triple of Lecture 20. Up to a scalar, Q^{-1} is the matrix c of Exercise 20.1.

Lemma 21.8. *For every $\theta \in \mathbb{R}$, the complex-linear extension of $\mathrm{Ad}(k_\theta)$ to $\mathfrak{g}_{\mathbb{C}}$ satisfies $\mathrm{Ad}(k_\theta)H' = H'$, $\mathrm{Ad}(k_\theta)E = e^{2i\theta}E$ and $\mathrm{Ad}(k_\theta)F = e^{-2i\theta}F$.*

Proof. Since $W = iH'$, we get

$$\mathrm{ad}(W)E = i[H', E] = 2iE, \qquad \mathrm{ad}(W)F = i[H', F] = -2iF, \qquad \mathrm{ad}(W)H' = 0.$$

By the relation between Ad and ad in Lecture 2, $\mathrm{Ad}(\exp \theta W) = e^{\theta \operatorname{ad} W}$, extended complex-linearly to $\mathfrak{g}_{\mathbb{C}}$. On an eigenvector of $\operatorname{ad} W$ with eigenvalue μ this operator is multiplication by $e^{\theta\mu}$. The eigenvalues 0, $2i$ and $-2i$ of $\operatorname{ad} W$ on H', E and F give the three formulas. ∎

Exercise 21.1 checks the adjoint lemma by matrix multiplication.

Proposition 21.9. *Let V be a $(\mathfrak{g}, K)$-module for $G = \mathrm{SL}(2, \mathbb{R})$ and $K = \mathrm{SO}(2)$. If $v \in V(n)$, then*

$$H' \cdot v = nv, \qquad E \cdot v \in V(n+2), \qquad F \cdot v \in V(n-2).$$

Proof. Let $v \in V(n)$. Since $\exp(\theta W) \cdot v = e^{in\theta} v$, the derivative condition (b) with $X = W$ gives

$$W \cdot v = \frac{d}{d\theta}\Big|_{\theta=0} e^{in\theta} v = in\, v, \qquad \text{so} \qquad H' \cdot v = -i\, W \cdot v = nv.$$

By the covariance condition (c), extended to $X \in \mathfrak{g}_{\mathbb{C}}$, and the adjoint lemma, Lemma 21.8,

$$k_\theta \cdot (E \cdot v) = (\mathrm{Ad}(k_\theta)E) \cdot (k_\theta \cdot v) = e^{2i\theta} E \cdot (e^{in\theta} v) = e^{i(n+2)\theta}\, E \cdot v.$$

So $E \cdot v \in V(n+2)$. The argument for F is the same with $e^{-2i\theta}$, and gives $F \cdot v \in V(n-2)$. ∎

In this proof the shift by 2 comes from the covariance condition (c). For this connected group K, the shift also follows from the derivative condition (b) and the bracket $[H', E] = 2E$. Indeed, this bracket and $H' \cdot v = nv$ show that H' multiplies $E \cdot v$ by $n + 2$. The operator H' acts on each summand $V(m)$ of $V = \bigoplus_{m \in \mathbb{Z}} V(m)$ by m, so $E \cdot v \in V(n+2)$. The bracket $[H', F] = -2F$ gives $F \cdot v \in V(n-2)$ in the same way.

Let Ω be the element

$$\Omega = H'^2 + 2EF + 2FE$$

of the universal enveloping algebra $U(\mathfrak{g}_{\mathbb{C}})$. It is the Casimir element of the triple. By the prerequisite Lie Algebras course, it is central in $U(\mathfrak{g}_{\mathbb{C}})$. Substituting $EF = FE + H'$ gives

$$\Omega = H'^2 + 2H' + 4FE, \qquad \Omega \cdot v = n(n+2)v + 4F \cdot E \cdot v \quad (v \in V(n)). \tag{21.2}$$

By the shift proposition, Proposition 21.9, $F \cdot E \cdot v \in V(n)$. So Ω preserves each $V(n)$.

Example 21.10. The group $\mathrm{SL}(2,\mathbb{R})$ acts on $\mathbb{C}^2$ by matrix multiplication, and hence on $V_m = \mathrm{Sym}^m \mathbb{C}^2$, of dimension $m+1$. The orbit maps are polynomial in the matrix entries, so every vector is smooth. Since V_m is finite-dimensional, every vector is also K-finite. Hence V_m is its own $(\mathfrak{g}, K)$-module. The Lie algebra acts on products by the Leibniz rule. For $u_\pm = (1, \pm i)^T$ we have

$$k_\theta u_+ = \begin{pmatrix} \cos\theta + i\sin\theta \\ -\sin\theta + i\cos\theta \end{pmatrix} = e^{i\theta} u_+, \qquad k_\theta u_- = e^{-i\theta} u_-.$$

Direct multiplication (Exercise 21.2) gives $Eu_- = u_+$, $Eu_+ = 0$, $Fu_+ = u_-$ and $Fu_- = 0$. Let $w_j = u_+^j u_-^{m-j}$ for $0 \le j \le m$. The group acts on each factor, so

$$k_\theta w_j = (e^{i\theta} u_+)^j (e^{-i\theta} u_-)^{m-j} = e^{i(2j-m)\theta} w_j.$$

Hence the K-types of V_m are $-m, -m+2, \dots, m$, each with multiplicity one. The Leibniz rule gives

$$E \cdot w_j = (m-j) w_{j+1} \quad (0 \le j < m), \qquad F \cdot w_j = j w_{j-1} \quad (0 < j \le m).$$

The endpoint actions are $Ew_m = 0$ and $Fw_0 = 0$, in agreement with the shift proposition, Proposition 21.9. Since $E \cdot w_m = 0$ and $w_m \in V(m)$, the Casimir formula (21.2) gives $\Omega \cdot w_m = m(m+2) w_m$. The K-types are the familiar $\mathfrak{sl}_2$-weights, read with respect to the compact element H' rather than the diagonal element H.

For $m \ge 1$ no inner product makes this module unitary. Indeed, $\mathrm{diag}(e^t, e^{-t})$ has the eigenvalue e^{mt} on V_m, and a unitary operator has eigenvalues of modulus one. Lecture 23 proves that every finite-dimensional continuous unitary representation of $\mathrm{SL}(2,\mathbb{R})$ is trivial. For this reason we allow nonunitary Π in this lecture.

If Π is unitary and Ω acts on $\mathcal{H}_K^\infty$ by a scalar λ, the skew-symmetry (F3) turns the Casimir formula (21.2) into formulas for $\|d\Pi(E)v\|^2$ and $\|d\Pi(F)v\|^2$. Exercise 21.4 derives them. Here $\langle d\Pi(E)u, w\rangle = -\langle u, d\Pi(F)w\rangle$ for smooth vectors u and w, because $F = \overline{E}$. In a unitary representation of $\mathrm{SU}(2)$ the same triple gives the sign +, because $F = E^*$ as matrices. In the physics notation $n = 2j$, the value $n(n+2)$ in (21.2) is $4j(j+1)$.

Exercise 21.3 builds infinite-dimensional $(\mathfrak{g}, K)$-modules from the same raising and lowering pattern. Its module V_λ has a basis v_n, $n \in 2\mathbb{Z}$, with $E \cdot v_n = v_{n+2}$ and $F \cdot v_n = \frac{1}{4}\big(\lambda - (n-2)n\big) v_{n-2}$. Figure 21.1 compares it with the module V_4 of Example 21.10. The shift proposition, Proposition 21.9, says that every arrow joins neighbouring dots of one parity. A missing arrow means that the operator gives zero. In the figure, generic means that $\lambda + 1$ is not the square of an odd integer. Exercise 21.3 shows that V_λ is then irreducible.

21.4 What the $(\mathfrak{g}, K)$-module does not record

A $(\mathfrak{g}, K)$-module has no topology. It does not record the norm or the action of elements of G outside K. We cite theorems that relate it back to representations. None of them is proved or used later in this lecture.

Admissibility. A *linear connected reductive group* is a closed connected group of real or complex matrices that is stable under conjugate transpose (Knapp RT, Section I.1). Examples are

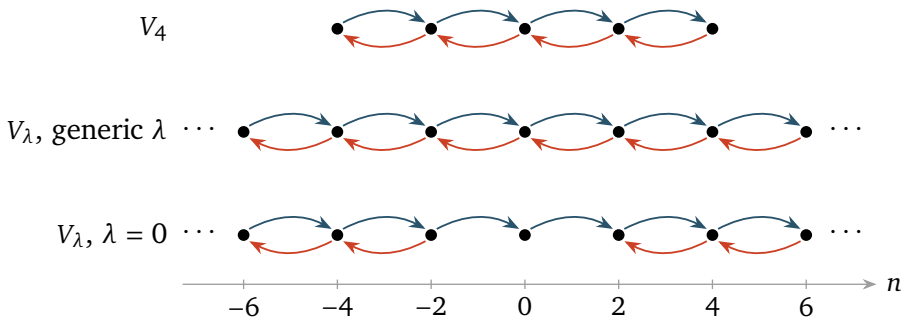


Figure 21.1: Ladders of K-types. Each dot is a one-dimensional space $V(n)$. Upper arrows show E and lower arrows show F. The rows show V_4, the module V_λ of Exercise 21.3 for generic λ, and the same family at $\lambda = 0$, where F vanishes on v_0 and v_2.

$\mathrm{SL}(2,\mathbb{R})$ and $\mathrm{SU}(p,q)$. This class is not the class of Lecture 20, as the note at the end of this lecture explains.

A *maximal compact subgroup* of a group is a compact subgroup contained in no strictly larger compact subgroup. For a linear connected reductive group $G \subseteq \mathrm{GL}(n,\mathbb{C})$, the subgroup $K = G \cap \mathrm{U}(n)$ is a maximal compact subgroup (Knapp RT, Section I.1, Proposition 1.2). For $G = \mathrm{SL}(2,\mathbb{R})$ it is $\mathrm{SO}(2)$. If G is also semisimple, then $\theta(X) = -X^*$ is a Cartan involution of $\mathfrak{g}$, and this K is the group K of the Cartan and Iwasawa theorem of Lecture 20.

Let G be a linear connected reductive group and $K = G \cap \mathrm{U}(n)$. The following two statements are in Knapp RT, Chapter VIII, and the note at the end of this lecture gives their numbers.

(i) If a strongly continuous Hilbert representation Π is admissible and $\Pi|_K$ is unitary, then every K-finite vector is smooth, so $\mathcal{H}_K^\infty = \mathcal{H}_K$.

(ii) (Harish-Chandra) Every irreducible unitary representation of G is admissible.

The circle example, Example 21.4, shows the conclusion of the smoothness statement (i) directly in the case $G = K$.

Globalization. Passing from a $(\mathfrak{g}, K)$-module to a continuous group representation is called globalization. A Harish-Chandra module is an admissible $(\mathfrak{g}, K)$-module finitely generated over $U(\mathfrak{g}_\mathbb{C})$. A positive definite Hermitian form on a Harish-Chandra module is invariant if the K-action is unitary and the $\mathfrak{g}$-action is skew-symmetric for real $X \in \mathfrak{g}$.

For linear connected reductive G with $K = G \cap \mathrm{U}(n)$, an irreducible Harish-Chandra module with an invariant positive definite Hermitian form has an irreducible unitary globalization. It is unique up to unitary equivalence. This theorem is cited from P. Etingof's MIT 18.757 notes, Theorem 7.5 and Proposition 7.1. The theorem supplies an action of G on a Hilbert space completing the module, together with its strong continuity. A completion of the vector space alone supplies neither.

One $(\mathfrak{g}, K)$-module can arise from completions of different kinds. Take $G = K = \mathrm{SO}(2)$ as in the circle example, Example 21.4. Translation also acts continuously on the Banach space $C(\mathbb{R}/2\pi\mathbb{Z})$ with the supremum norm, which is not a Hilbert space. With the same definitions, the smooth K-finite vectors of $C(\mathbb{R}/2\pi\mathbb{Z})$ are again the trigonometric polynomials.

So $L^2(\mathbb{R}/2\pi\mathbb{Z})$ and $C(\mathbb{R}/2\pi\mathbb{Z})$ have the same $(\mathfrak{g}, K)$-module. So the module does not record the norm.

This common module has infinitely many distinct circle types and is not finitely generated over $U(\mathfrak{k}_{\mathbb{C}})$. Thus this example is outside the irreducible Harish-Chandra theorem. The more general Casselman–Wallach globalization theorem belongs to further study.

Integration. The identities among $d\Pi(E)$, $d\Pi(F)$ and $d\Pi(H')$ were valid because Π was given. Formal commutation relations among unbounded operators do not by themselves produce a group representation. For the modules of Exercise 21.3, the question of which ones come from unitary representations is taken up in Lecture 22 and Lecture 23.

21.5 Self-checks

Check 1. In Step 3 of the proof of the $(\mathfrak{g}, K)$-module proposition, Proposition 21.6, why must all of W_v consist of smooth vectors?

Answer. The space U is spanned by vectors $d\Pi(X_i)w_j$ with $w_j \in W_v$. These are defined only for smooth w_j. Step 1 shows that each $\Pi(k)v$ is smooth.

Check 2. What are the K-types of Ad on $\mathfrak{g}_{\mathbb{C}}$? Which V_m has the same K-types?

Answer. By the adjoint lemma, Lemma 21.8, the elements E, H' and F have K-types 2, 0, -2, each with multiplicity one. The module V_2 has the same K-types.

Check 3. Could the eigenvalues of $H = \operatorname{diag}(1, -1)$ serve as K-type labels?

Answer. No. The derivative condition (b) relates the K-action only to elements of $\mathfrak{k}_{\mathbb{C}} = \mathbb{C}H'$. Since $H \notin \mathfrak{k}_{\mathbb{C}}$, the action of H is not determined by the K-action. In the V_m example, Example 21.10, the element $H = E + F$ moves $w_j \in V(n)$, where $n = 2j - m$, into $V(n+2) \oplus V(n-2)$.

Summary

Let K be a compact subgroup of G. For a strongly continuous representation of G on a Hilbert space, the smooth K-finite vectors carry an action of $\mathfrak{g}$ and of K. The derivative condition (b) relates the two actions on $\mathfrak{k}$. The covariance condition (c) describes how the K-action transforms the $\mathfrak{g}$-action. The proof that $\mathfrak{g}$ preserves K-finiteness uses the covariance condition.

For the circle, the K-finite vectors are the trigonometric polynomials. For $\mathrm{SL}(2, \mathbb{R})$ with $K = \mathrm{SO}(2)$, the covariance condition (c) shows that E and F shift the circle K-types by 2 and -2, respectively. The derivative condition (b) shows that H' acts on the K-type n by multiplication by n. For this connected K, the brackets $[H', E] = 2E$ and $[H', F] = -2F$ then give the same shifts.

For linear connected reductive G with $K = G \cap \mathrm{U}(n)$, three results are cited. Irreducible unitary representations are admissible. In an admissible strongly continuous Hilbert representation Π with $\Pi|_K$ unitary, K-finite vectors are smooth. An irreducible Harish-Chandra module

with an invariant positive definite Hermitian form has an irreducible unitary globalization, unique up to unitary equivalence.

Sources and hypotheses

The class of groups. Knapp RT, Section I.1, defines linear connected reductive groups, and calls such a group with finite centre linear connected semisimple. Its Proposition 1.2 states that K is compact, connected and maximal compact, and that $K \times \mathfrak{p} \to G$ is a diffeomorphism. This class differs from the class of Lecture 20, which consists of connected semisimple Lie groups with finite centre. For example, the identity component $\mathrm{GL}(n, \mathbb{R})_0$ lies in the present class and is not semisimple, whereas a nonlinear finite cover of $\mathrm{SL}(2, \mathbb{R})$ lies only in the class of Lecture 20. The group $\mathrm{SL}(2, \mathbb{R})$ lies in both.

Admissibility. The smoothness statement (i) is Knapp RT, Proposition 8.5. The definition immediately after the proof of Theorem 8.1 includes the unitarity of $\Pi|_K$ and finite multiplicities, which are the hypotheses of the smoothness statement. The admissibility statement (ii) is Knapp RT, Theorem 8.1.

Globalization. In the notes of Etingof, the unitary globalization is Theorem 7.5, printed p. 38, and its uniqueness is Proposition 7.1, printed p. 37.

Reading

- Knapp RT, Section I.1: linear connected reductive groups.
- Knapp RT, Sections III.3–4: smooth vectors and the facts (F1)–(F3).
- Knapp RT, Chapter VIII: admissible representations and Harish-Chandra's admissibility theorem.
- P. Etingof, MIT 18.757 *Representations of Lie Groups*, lecture notes, Lectures 2 and 5–7: a second account of this material.
- Etingof, Lecture 7, Theorem 7.5 (printed p. 38) and Proposition 7.1 (printed p. 37): the unitary globalization result stated here and its uniqueness. For further study, Wallach, *Real Reductive Groups II*, develops Casselman–Wallach globalization.